

End-to-End Vendor Performance Analytics using Python, SQLite, and Power BI

CHAPTER 1

1.1 Introduction

- In today's competitive business environment, organizations rely on multiple vendors to procure products and maintain an efficient supply chain. Evaluating vendor performance is essential for improving procurement strategies, controlling inventory costs, increasing profitability, and ensuring smooth business operations. By analyzing vendor sales, purchase patterns, and profitability, businesses can identify high-performing vendors, detect underperforming suppliers, and make informed decisions based on data rather than intuition.
- This project presents an end-to-end Vendor Performance Analysis solution developed using Python, SQLite, and Microsoft Power BI. The project begins with a Python-based ETL (Extract, Transform, Load) pipeline that extracts data from multiple CSV files, transforms the raw data through cleaning and integration, and loads the processed data into a SQLite database. During the transformation stage, multiple datasets were merged to create a consolidated vendor summary table, and important business metrics such as Gross Profit, Profit Margin, Stock Turnover, Sales-to-Purchase Ratio, Purchase Contribution, and Unsold Capital were calculated to support business analysis.
- After completing the ETL process, Exploratory Data Analysis (EDA) was performed using Python to understand vendor behaviour, sales trends, inventory movement, and profitability. The processed dataset was then imported into Microsoft Power BI, where an interactive dashboard was designed using DAX measures, KPI cards, slicers, and multiple visualizations. The dashboard enables users to monitor vendor performance, compare sales and purchases, identify top-performing products, analyze purchase contribution, and evaluate business profitability through dynamic and interactive reports.
- The developed solution demonstrates how data engineering, data analysis, and business intelligence can be integrated into a single workflow to transform raw business data into meaningful insights. The dashboard supports data-driven decision-making and provides valuable information that can help organizations optimize vendor relationships, improve inventory planning, reduce unsold stock, and enhance overall business performance.

1.2 Problem Statement

- Organizations often procure products from numerous vendors, making it difficult to monitor vendor performance using traditional spreadsheet-based analysis. Without a centralized analytical solution, businesses struggle to identify profitable vendors, evaluate inventory movement, compare purchase and sales trends, and detect underperforming suppliers. Manual analysis is time-consuming, error-prone, and unable to provide interactive insights required for effective decision-making.

- The objective of this project is to develop an end-to-end vendor performance analytics solution by implementing a Python-based ETL pipeline, performing exploratory data analysis, calculating key business performance indicators, and designing an interactive Power BI dashboard. The solution aims to provide meaningful insights into vendor performance, product sales, profitability, and inventory efficiency, enabling organizations to make informed business decisions.

1.3 Objectives

The primary objectives of this project are:

- To develop a Python-based ETL pipeline for extracting, transforming, and loading vendor sales data from multiple CSV files.
- To clean, integrate, and preprocess the raw data using Python and SQLite.
- To perform Exploratory Data Analysis (EDA) to identify sales trends, purchasing behaviour, inventory movement, and vendor performance.
- To calculate important business KPIs such as Total Sales, Total Purchase, Gross Profit, Profit Margin, Purchase Contribution, Unsold Capital, Stock Turnover, and Sales-to-Purchase Ratio.
- To identify top-performing and low-performing vendors based on business metrics.
- To analyse product-wise sales performance and vendor contribution.
- To design an interactive Power BI dashboard using DAX measures, KPI cards, slicers, and analytical visualizations.
- To generate actionable business insights that support vendor management, inventory optimization, and data-driven decision-making.

CHAPTER 2

Project Architecture and System Overview

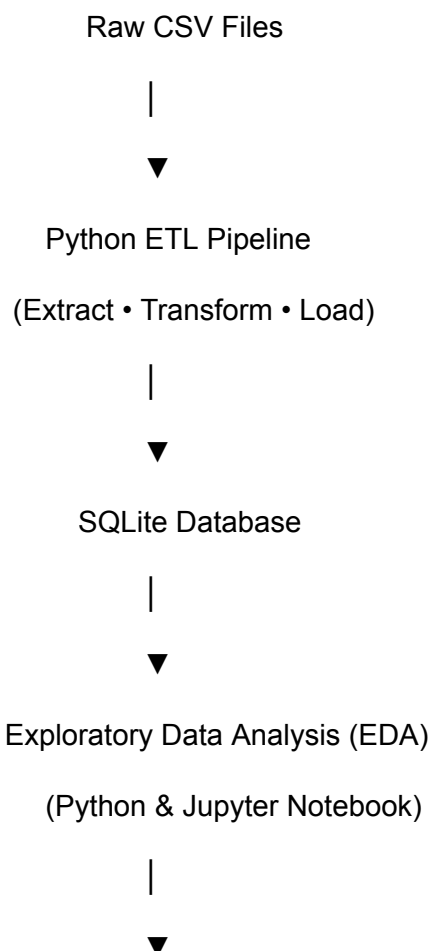
2.1 Project Overview

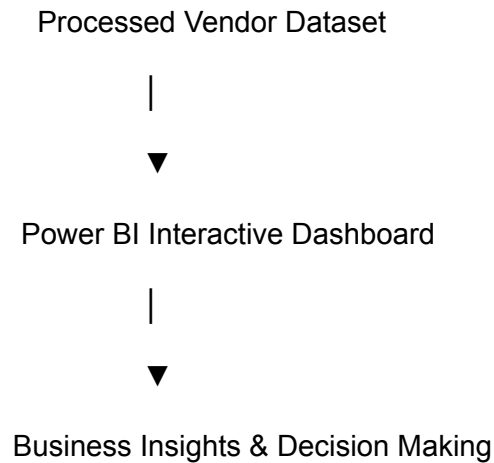
The Vendor Performance Analysis project is designed as an end-to-end data analytics solution that integrates data engineering, data analysis, and business intelligence. The project begins with raw vendor transaction data stored in multiple CSV files. A Python-based ETL pipeline extracts the data, performs preprocessing and transformation operations, and loads the processed data into a SQLite database. The transformed data is then analyzed using Python, and the final processed dataset is visualized through an interactive Power BI dashboard.

The project follows a structured workflow to ensure that raw business data is converted into meaningful insights. Each stage of the workflow contributes to improving data quality, calculating important business metrics, and generating interactive reports that support informed decision-making.

2.2 System Architecture

The overall architecture of the project consists of six major stages.





The architecture demonstrates how raw transactional data is transformed into meaningful business insights using modern data analytics tools.

2.3 Project Workflow

The project was implemented in the following phases:

Phase 1 – Data Collection

The project utilizes multiple CSV files containing vendor purchase records, sales transactions, inventory information, freight details, and product pricing information.

Phase 2 – ETL Pipeline

A Python-based ETL pipeline was developed to automate data processing.

During this phase:

- Multiple CSV files were extracted.
- Large datasets were processed efficiently using chunk-based ingestion.
- Data from different sources was merged to create a consolidated vendor summary.
- Logging was implemented to monitor each processing step and record successful execution.
- The processed data was loaded into a SQLite database for further analysis.

Phase 3 – Data Cleaning and Transformation

After extracting the data, several preprocessing operations were performed to improve data quality.

These operations included:

- Handling missing values.
- Correcting data types.
- Standardizing text values.

- Removing inconsistencies.
- Preparing the dataset for analysis.

Additional business metrics were also created, including:

- Gross Profit
- Profit Margin
- Stock Turnover
- Sales-to-Purchase Ratio
- Purchase Contribution
- Unsold Capital

Phase 4 – Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed using Python to understand the characteristics of the data before visualization.

The analysis focused on:

- Vendor performance
- Product sales
- Purchase behaviour
- Profitability
- Inventory movement
- Sales trends

The insights obtained during EDA helped identify important business patterns and supported the design of the final dashboard.

Phase 5 – Dashboard Development

The processed dataset was imported into Microsoft Power BI.

An interactive dashboard was developed using:

- DAX Measures
- KPI Cards
- Dynamic Slicers
- Clustered Bar Charts
- Clustered Column Charts
- Scatter Plot
- Donut Chart

The dashboard allows users to interactively analyze vendor performance and business metrics.

Phase 6 – Business Insights

The final stage of the project focuses on converting analytical results into business insights.

The dashboard enables organizations to:

- Identify top-performing vendors.
- Detect low-performing vendors.
- Monitor product sales performance.
- Evaluate vendor contribution.
- Analyze profitability.
- Support inventory and procurement decisions.

2.4 Technologies Used

The following technologies were used throughout the project.

Category	Technology
Programming Language	Python
Data Processing	Pandas, NumPy
Data Visualization (EDA)	Matplotlib, Seaborn
Database	SQLite
Development Environment	Jupyter Notebook
Business Intelligence	Microsoft Power BI Desktop
Dashboard Calculations	DAX (Data Analysis Expressions)
Logging	Python Logging Module
Data Storage	CSV Files

2.5 Project Deliverables

The final outcomes of the project include:

- Python-based ETL Pipeline
- SQLite Database containing processed vendor data
- Exploratory Data Analysis (EDA)
- Cleaned Vendor Sales Dataset
- Interactive Power BI Dashboard
- Business Performance Insights
- Technical Project Report

CHAPTER 3

Dataset Description and Data Preparation

3.1 Dataset Overview

The Vendor Performance Analysis project uses multiple datasets containing information related to vendor purchases, product sales, inventory, product pricing, freight charges, and vendor invoices. Instead of relying on a single data source, the project integrates several CSV files to generate a consolidated vendor performance dataset.

The raw datasets contain transactional records collected from different stages of the business process. Since the information is distributed across multiple files, an ETL (Extract, Transform, Load) pipeline was developed to combine these datasets into a unified vendor summary table for analysis.

The final processed dataset serves as the primary source for Exploratory Data Analysis (EDA) and the Power BI dashboard.

3.2 Source Datasets

The project utilizes the following CSV files:

Dataset	Description
begin_inventory.csv v	Contains the opening inventory details of products.
end_inventory.csv	Contains the closing inventory details after sales transactions.
purchases.csv	Stores vendor purchase transactions, including purchased quantities and purchase amounts.
purchase_prices.csv v	Contains product purchase prices and pricing information.

sales.csv Contains product sales transactions, sales quantities, and sales revenue.

vendor_invoice.csv Stores vendor invoice and freight-related information.

Each dataset contributes different business information required to perform a comprehensive vendor performance analysis.

3.3 Data Integration

Since business information was distributed across multiple datasets, the individual CSV files could not be analyzed independently.

The ETL pipeline integrated the datasets by combining purchase, sales, pricing, inventory, and freight information into a single consolidated vendor summary table. SQL queries and Python were used to merge the datasets based on common business attributes such as Vendor Number, Brand, and Product Description.

The consolidated dataset provides a complete view of vendor performance by combining financial, inventory, and sales-related information into one table.

3.4 Data Cleaning

Before performing analysis, several preprocessing steps were carried out to improve data quality and ensure consistency.

The major data cleaning operations included:

- Handling missing values.
- Removing duplicate records where necessary.
- Correcting inconsistent data types.
- Standardizing text values.
- Cleaning vendor and product information.
- Preparing numerical columns for analysis.
- Verifying data consistency before loading into the database.

These preprocessing steps improved the reliability and accuracy of the subsequent analysis.

3.5 Derived Business Metrics

In addition to the original dataset attributes, several new business metrics were created to provide deeper insights into vendor performance.

The calculated metrics include:

- Gross Profit

- Profit Margin
- Stock Turnover
- Sales-to-Purchase Ratio
- Purchase Contribution
- Unsold Capital

These metrics were generated during the transformation stage of the ETL pipeline and later used as Key Performance Indicators (KPIs) in the Power BI dashboard.

3.6 Final Dataset

After completing the ETL process and data cleaning, a consolidated vendor summary dataset was generated.

The final dataset contains all the information required for vendor performance analysis, including:

- Vendor Information
- Product Information
- Purchase Details
- Sales Details
- Inventory Information
- Freight Cost
- Gross Profit
- Profit Margin
- Stock Turnover
- Sales-to-Purchase Ratio

This processed dataset was exported and used as the primary input for Exploratory Data Analysis and Power BI dashboard development.

3.7 Importance of Data Preparation

Data preparation is one of the most critical stages of any data analytics project. Business decisions based on incomplete or inconsistent data may lead to inaccurate conclusions.

By integrating multiple datasets, cleaning the raw data, and generating meaningful business metrics, the project ensured that the analytical results and dashboard visualizations accurately represented vendor performance.

The prepared dataset provided a reliable foundation for identifying high-performing vendors, evaluating profitability, analyzing inventory movement, and supporting data-driven business decisions.

CHAPTER 4

ETL Pipeline Implementation

4.1 Introduction

To ensure efficient processing of the vendor datasets, an ETL (Extract, Transform, Load) pipeline was developed using Python. Since the project data was distributed across multiple CSV files, the ETL pipeline automated the process of reading, integrating, cleaning, transforming, and storing the data in a structured format.

The ETL pipeline served as the foundation of the project by preparing a clean and consolidated dataset that could be used for exploratory data analysis and Power BI dashboard development.

4.2 ETL Workflow

The ETL pipeline follows three major stages:

Extract



Transform



Load

Each stage is explained below.

4.3 Extract Phase

During the extraction phase, the raw data was collected from multiple CSV files containing vendor purchases, product sales, inventory details, purchase prices, freight information, and vendor invoices.

The extraction process was automated using Python. Large CSV files were processed in chunks to improve memory efficiency and reduce processing time.

The extracted datasets include:

- begin_inventory.csv
- end_inventory.csv
- purchases.csv
- purchase_prices.csv

- sales.csv
- vendor_invoice.csv

Processing each dataset individually allowed the pipeline to efficiently handle large volumes of transactional data before combining them into a unified dataset.

4.4 Transform Phase

The transformation stage converts raw business data into a structured and analysis-ready format.

During this stage, the following operations were performed:

Data Integration

Multiple datasets were merged using SQL queries and Python to create a consolidated Vendor Summary table.

The integration combined:

- Purchase Information
- Sales Information
- Inventory Details
- Product Pricing
- Freight Cost

into a single dataset.

Data Cleaning

To improve data quality, several preprocessing operations were performed:

- Handling missing values.
- Correcting data types.
- Standardizing text fields.
- Removing inconsistencies.
- Preparing numerical columns for calculations.

These operations ensured that the final dataset was suitable for analysis and visualization.

Business Metric Calculation

After cleaning the data, several business metrics were generated to evaluate vendor performance.

The calculated metrics include:

- Gross Profit
- Profit Margin
- Stock Turnover

- Sales-to-Purchase Ratio

These metrics provide meaningful insights into vendor profitability, inventory efficiency, and sales performance.

4.5 Load Phase

Once the transformation process was completed, the processed vendor summary dataset was loaded into a SQLite database.

The final Vendor Summary table served as the central data source for subsequent analysis.

The processed dataset was then exported and imported into Microsoft Power BI for dashboard development.

Loading the transformed data into a structured database improved data organization and simplified the analytical workflow.

4.6 Logging Mechanism

To improve the reliability and traceability of the ETL pipeline, Python's logging module was implemented.

The logging mechanism records:

- Start of data processing.
- Processing of individual CSV files.
- Chunk-wise data insertion.
- Successful completion of data ingestion.
- Vendor summary generation.
- Data cleaning status.
- Completion of the loading process.

The log files also record execution status and processing time, making it easier to monitor the ETL pipeline and identify potential issues during execution.

4.7 Advantages of the ETL Pipeline

Implementing the ETL pipeline provided several benefits:

- Automated processing of multiple datasets.
- Reduced manual effort in data preparation.
- Improved data consistency.
- Better handling of large datasets through chunk processing.
- Centralized storage of processed data in SQLite.
- Reusable workflow for future data updates.
- Reliable execution monitoring through log files.
-

CHAPTER 5

Feature Engineering and Business Metrics

5.1 Introduction

Feature Engineering is the process of creating new variables from existing data to improve analysis and generate meaningful business insights. The original vendor datasets contained transactional information such as purchase quantities, sales quantities, purchase prices, sales prices, and freight costs. However, these raw values alone were not sufficient to evaluate vendor performance effectively.

To obtain meaningful insights, several business metrics were derived during the transformation stage of the ETL pipeline. These engineered features were later used for Exploratory Data Analysis (EDA), KPI calculation, and Power BI dashboard development.

5.2 Gross Profit

Gross Profit is one of the most important financial indicators used to measure the profitability of a product or vendor.

It represents the difference between total sales revenue and total purchase cost.

Formula

Gross Profit = Total Sales Dollars – Total Purchase Dollars

Business Significance

- Measures the actual profit generated from product sales.
- Helps identify profitable vendors.
- Supports pricing and purchasing decisions.
- Used as a Key Performance Indicator (KPI) in the Power BI dashboard.

5.3 Profit Margin

Profit Margin indicates the percentage of revenue retained as profit after deducting purchase costs.

Unlike Gross Profit, Profit Margin provides a relative measure of profitability, making it easier to compare vendors of different sizes.

Formula

Profit Margin (%) = (Gross Profit / Total Sales Dollars) × 100

Business Significance

- Evaluates vendor profitability.

- Identifies low-margin products.
- Supports pricing strategy.
- Helps compare vendor performance objectively.

5.4 Stock Turnover

Stock Turnover measures how efficiently inventory is sold over a given period.

A higher Stock Turnover indicates efficient inventory movement, whereas a lower value may indicate excess inventory or slow-moving products.

Business Significance

- Measures inventory efficiency.
- Identifies slow-moving products.
- Supports inventory planning.
- Reduces storage costs.

5.5 Sales-to-Purchase Ratio

The Sales-to-Purchase Ratio compares the quantity sold with the quantity purchased.

This metric helps determine whether inventory is being converted into sales efficiently.

Business Significance

- Evaluates purchasing effectiveness.
- Identifies overstocked products.
- Helps maintain optimum inventory levels.
- Supports procurement decisions.

5.6 Purchase Contribution

Purchase Contribution represents the percentage contribution of each vendor toward the total purchase value.

This metric helps identify vendors responsible for the majority of procurement expenditure.

Business Significance

- Identifies major vendors.
- Supports vendor prioritization.
- Helps analyze supplier dependency.
- Used in the Purchase Contribution donut chart.

5.7 Unsold Capital

Unsold Capital represents the amount of capital invested in inventory that has not yet been converted into sales.

It provides an estimate of funds blocked in unsold stock.

Business Significance

- Measures capital tied up in inventory.
- Identifies excess stock.
- Supports inventory optimization.
- Helps reduce financial risk.

5.8 Importance of Feature Engineering

The engineered business metrics significantly improved the analytical capability of the project. Instead of relying only on raw transactional data, these features enabled a comprehensive evaluation of vendor performance, profitability, inventory efficiency, and purchasing behaviour.

These engineered metrics also served as the foundation for the Key Performance Indicators (KPIs) and visualizations developed in the Power BI dashboard, allowing business users to make informed decisions through interactive reporting.

5.9 Summary

Feature Engineering transformed raw transactional data into meaningful business indicators that could be easily interpreted by decision-makers. Metrics such as Gross Profit, Profit Margin, Stock Turnover, Sales-to-Purchase Ratio, Purchase Contribution, and Unsold Capital provided deeper insights into vendor performance and played a vital role in the overall success of the Vendor Performance Analysis project.

CHAPTER 6

Exploratory Data Analysis (EDA)

6.1 Introduction

Exploratory Data Analysis (EDA) is an important stage in any data analytics project that helps in understanding the structure, quality, and characteristics of the dataset before performing detailed business analysis. The objective of EDA is to examine the available data, identify inconsistencies, validate data quality, and ensure that the dataset is suitable for further processing.

In this project, EDA was performed after completing the ETL pipeline. The consolidated vendor summary dataset stored in SQLite was loaded into Jupyter Notebook for initial exploration and validation. The analysis focused on understanding the dataset structure, verifying data quality, examining descriptive statistics, and preparing the data for feature engineering and business analysis.

6.2 Objectives of Exploratory Data Analysis

The primary objectives of EDA in this project were:

- To understand the structure of the processed dataset.
- To verify the quality of the integrated data.
- To inspect numerical and categorical attributes.
- To identify missing values and inconsistencies.
- To validate the derived business metrics.
- To prepare the dataset for business analysis and dashboard development.

6.3 Dataset Exploration

The processed vendor summary dataset was initially explored to gain an understanding of its structure and contents.

The following exploratory operations were performed:

- Examining the first few records of the dataset.
- Checking the number of rows and columns.
- Understanding the available attributes.
- Reviewing data types of each column.
- Verifying the successful integration of purchase, sales, inventory, and freight information.

These steps ensured that the ETL process had generated a complete and consistent dataset for analysis.

6.4 Data Quality Assessment

To ensure the reliability of the analysis, the dataset was validated using several quality checks.

The following checks were performed:

- Missing value analysis.
- Duplicate record verification.
- Data type validation.
- Numerical consistency checks.
- Text standardization verification.

These validation steps confirmed that the processed dataset was suitable for further analytical processing.

6.5 Descriptive Statistics

Descriptive statistical analysis was performed to understand the distribution and characteristics of numerical variables.

The analysis included summary statistics for important attributes such as:

- Purchase Price
- Actual Selling Price
- Total Purchase Dollars
- Total Sales Dollars
- Gross Profit
- Profit Margin
- Stock Turnover

The statistical summary provided insights into the central tendency, variation, and range of the business data, helping identify any unusual values before conducting business analysis.

6.6 Validation of Engineered Features

After feature engineering, the newly created business metrics were examined to ensure that the calculations were accurate and meaningful.

The following engineered features were validated:

- Gross Profit
- Profit Margin
- Stock Turnover
- Sales-to-Purchase Ratio

The successful validation of these metrics ensured that they could be reliably used in subsequent analysis and dashboard development. These engineered features were later incorporated into Power BI as KPIs and analytical measures.

6.7 Key Observations

The exploratory analysis resulted in the following observations:

- The consolidated dataset successfully integrated information from multiple business sources.
- The processed data was complete and consistent after the ETL and cleaning stages.
- The engineered business metrics were successfully calculated and validated.
- The dataset was well-structured and ready for detailed vendor performance analysis and dashboard development.

6.8 Summary

Exploratory Data Analysis served as an intermediate validation stage between the ETL pipeline and business analysis. Rather than generating business insights, this phase focused on understanding the processed dataset, verifying data quality, and ensuring that the engineered features were suitable for further analytical processing.

CHAPTER 7

Vendor Performance Analysis and Business Insights

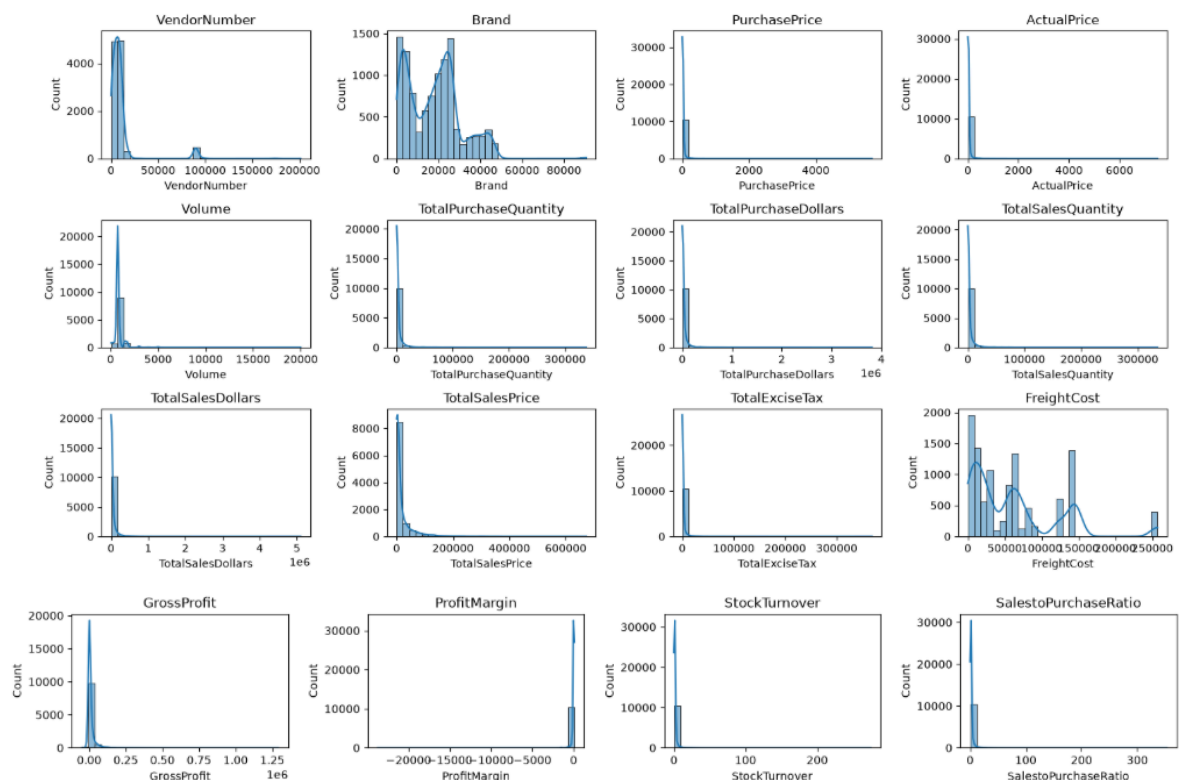
7.1 Introduction

After preparing the dataset through the ETL pipeline and feature engineering, a detailed analysis was performed to uncover meaningful business insights. Using Python libraries such as Pandas, Matplotlib, and Seaborn, multiple visualizations were created to examine data distribution, detect outliers, study relationships between variables, and evaluate vendor performance.

The objective of this analysis is to answer key business questions related to sales, purchases, profitability, inventory management, and vendor contribution. The insights obtained from this analysis also guided the design of the interactive Power BI dashboard developed in the final stage of the project.

In this phase of EDA, we will analyze the resultant table to gain insights into the distribution of each column. This will help us understand data patterns, identify anomalies, and ensure data quality before proceeding with further analysis.

Figure 7.1: Distribution of Numerical Variables



Purpose

- To examine the distribution of all numerical variables in the vendor dataset.
- To identify data patterns such as skewness, concentration, and variability.
- To gain an initial understanding of the dataset before performing advanced statistical and business analyses.

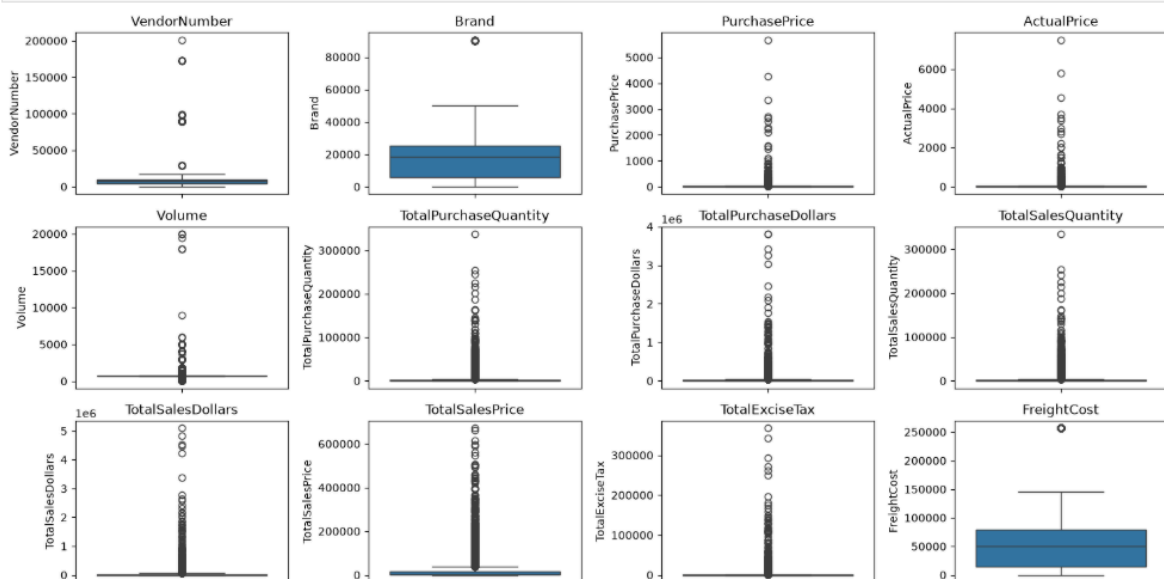
Key Observations

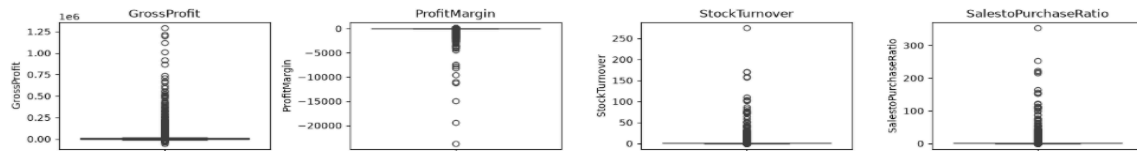
- Most financial variables, including Purchase Price, Actual Price, Total Purchase Dollars, Total Sales Dollars, and Gross Profit, exhibit a right-skewed distribution, indicating that lower-value transactions are more common than high-value transactions.
- Freight Cost shows multiple peaks, suggesting different freight charge ranges across vendors or product categories.
- Profit Margin, Stock Turnover, and Sales-to-Purchase Ratio display comparatively concentrated distributions, indicating more consistent values across the dataset.
- No abnormal distribution patterns were observed that would prevent further analysis.

Insights

- The dataset is dominated by lower-value transactions, while a relatively small number of transactions contribute significantly to overall sales and purchases.
- The skewed distributions indicate that a small number of vendors or products may account for a large portion of business activity.
- The distribution analysis provides confidence that the processed dataset is suitable for subsequent analyses, including outlier detection, correlation analysis, and vendor performance evaluation.

Figure 7.2: Box Plots of Numerical Variables





Purpose

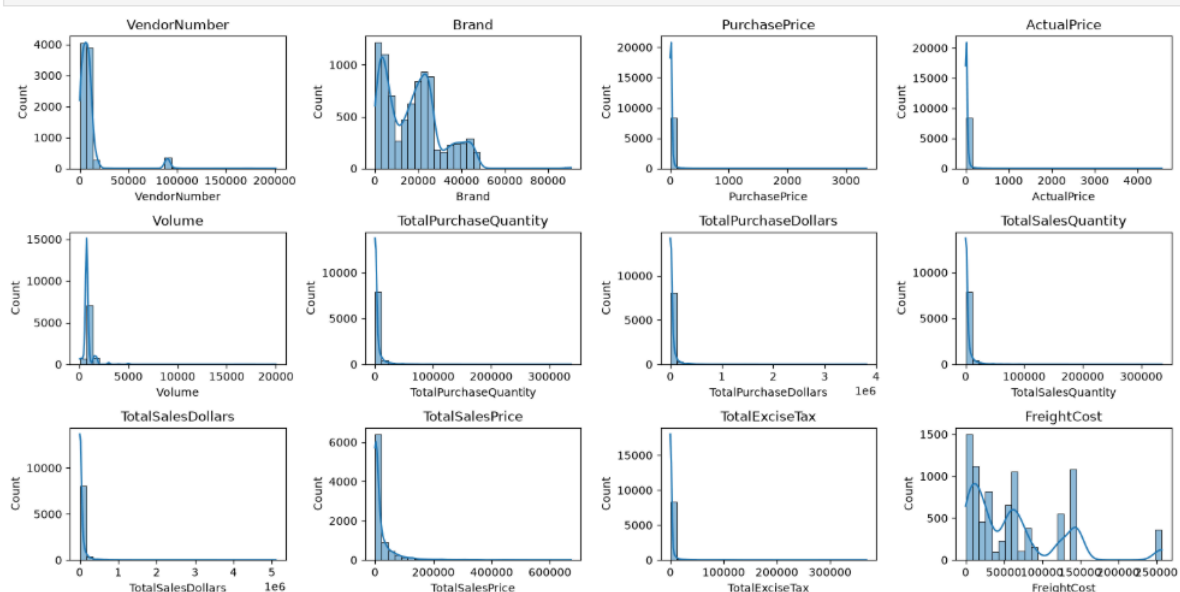
- To identify the presence of outliers in the numerical variables.
- To understand the spread and variability of the dataset.
- To detect extreme values that may influence business analysis.

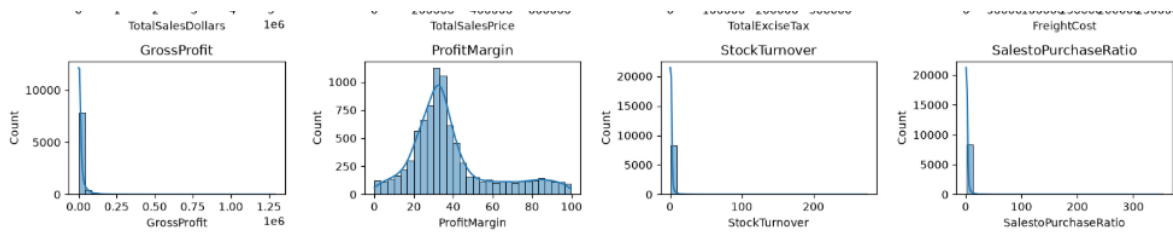
Key Observations

- Most numerical variables contain a considerable number of outliers beyond the upper whisker.
- Variables such as Purchase Price, Actual Price, Total Purchase Dollars, Total Sales Dollars, and Gross Profit exhibit several high-value outliers.
- Profit Margin contains a few negative outliers, indicating transactions with exceptionally low profitability.
- Stock Turnover and Sales-to-Purchase Ratio also show extreme values for a limited number of records.

Key Insights

- The presence of outliers suggests that a small number of vendors or products contribute disproportionately to sales, purchases, and profits.
- These outliers represent genuine business scenarios, such as high-value transactions or premium products, and were retained for further analysis rather than being removed.
- Understanding these extreme values is essential for identifying high-performing vendors and evaluating exceptional business cases.
- **Figure 7.3: Distribution of Numerical Variables After Handling Inconsistencies**





Data Preprocessing Summary

- Handled missing values.
- Standardized data types.
- Corrected inconsistent records.
- Validated engineered business metrics.
- Prepared the dataset for statistical analysis.

Purpose

- To verify the data distribution after handling inconsistencies and preprocessing.
- To ensure that the cleaned dataset is suitable for further analysis and visualization.
- To validate the effectiveness of the data preparation process.

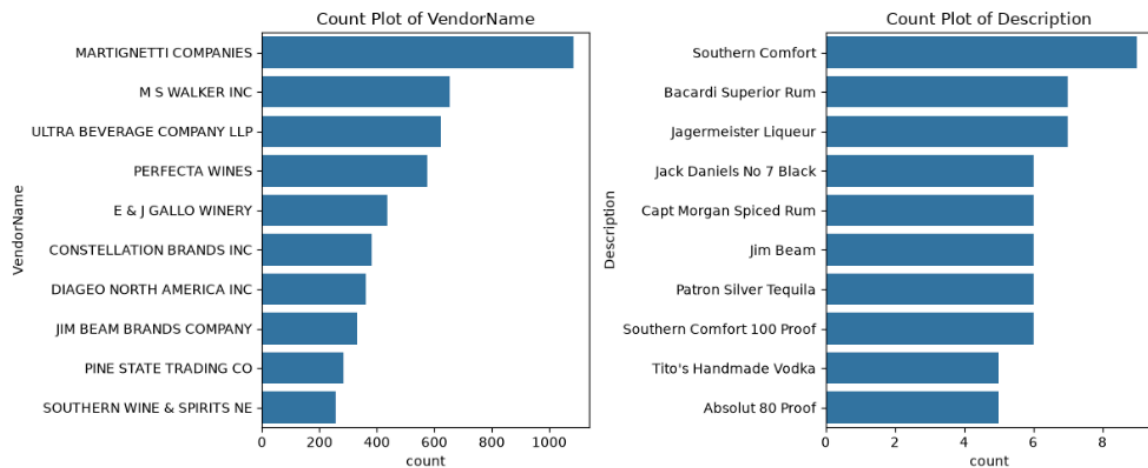
Key Observations

- The overall distribution of the numerical variables remains consistent after preprocessing.
- Most variables continue to exhibit a right-skewed distribution, which is expected for transactional business data.
- The distributions appear cleaner and more consistent, indicating successful handling of data inconsistencies.

Key Insights

- Data preprocessing improved the overall quality of the dataset without altering the underlying business patterns.
- The cleaned dataset is reliable for subsequent statistical analysis, feature engineering, and dashboard development.
- Preserving the natural distribution ensures that the business insights derived from the analysis remain realistic and meaningful.

Figure 7.4: Count Plot of Vendor Names and Product Descriptions



Purpose

- To identify the most frequently occurring vendors and product descriptions in the dataset.
- To understand the distribution of vendor records and product entries.
- To gain an overview of the dataset composition before performing detailed performance analysis.

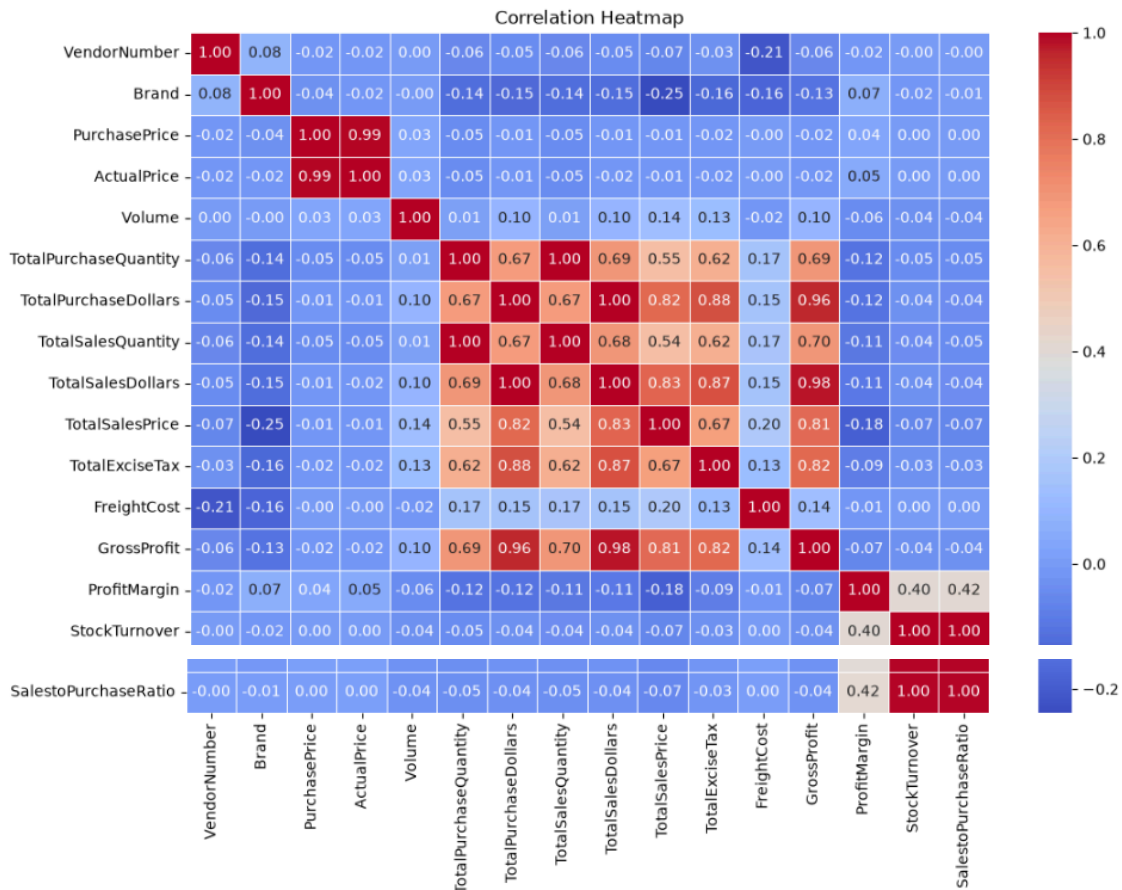
Key Observations

- MARTIGNETTI COMPANIES has the highest number of records among all vendors, followed by M S WALKER INC and ULTRA BEVERAGE COMPANY LLP.
- Among product descriptions, Southern Comfort appears most frequently, followed by Bacardi Superior Rum and Jägermeister Liqueur.
- The distribution indicates that certain vendors and products have significantly higher representation in the dataset than others.

Key Insights

- Vendors with a higher number of records are not necessarily the highest-performing vendors; further analysis using sales and profit metrics is required to evaluate their business performance.
- Frequently occurring product descriptions indicate products with higher transaction activity, making them important candidates for detailed sales and inventory analysis.

Figure 7.5: Correlation Heatmap of Numerical Variables



Purpose

- To analyze the relationships between numerical variables in the dataset.
- To identify strong positive or negative correlations among business metrics.
- To understand which variables have the greatest influence on vendor performance.

Key Observations

- Purchase Price and Actual Price show a very strong positive correlation (0.99), indicating that products with higher purchase costs generally have higher selling prices.
- Gross Profit has a strong positive correlation with Total Sales Dollars (0.98) and Total Purchase Dollars (0.96), highlighting that higher sales and purchases contribute significantly to profitability.
- Total Purchase Dollars, Total Sales Dollars, Total Sales Price, and Total Excise Tax are strongly correlated, reflecting their close financial relationship.
- Profit Margin shows a moderate positive correlation with Stock Turnover (0.40) and Sales-to-Purchase Ratio (0.42), suggesting that efficient inventory movement contributes to improved profitability.
- Vendor Number and Brand exhibit weak correlations with most numerical variables, indicating that they primarily serve as identifier fields.

Key Insights

- Sales- and purchase-related metrics are the primary drivers of vendor profitability.
- Strong correlations among financial variables validate the consistency of the processed dataset and reflect expected business behavior.
- The correlation analysis helps identify the most influential metrics for evaluating vendor performance and supports data-driven business decision-making.

Research Question 1

Which brands require promotional or pricing adjustments due to lower sales performance despite having higher profit margins?

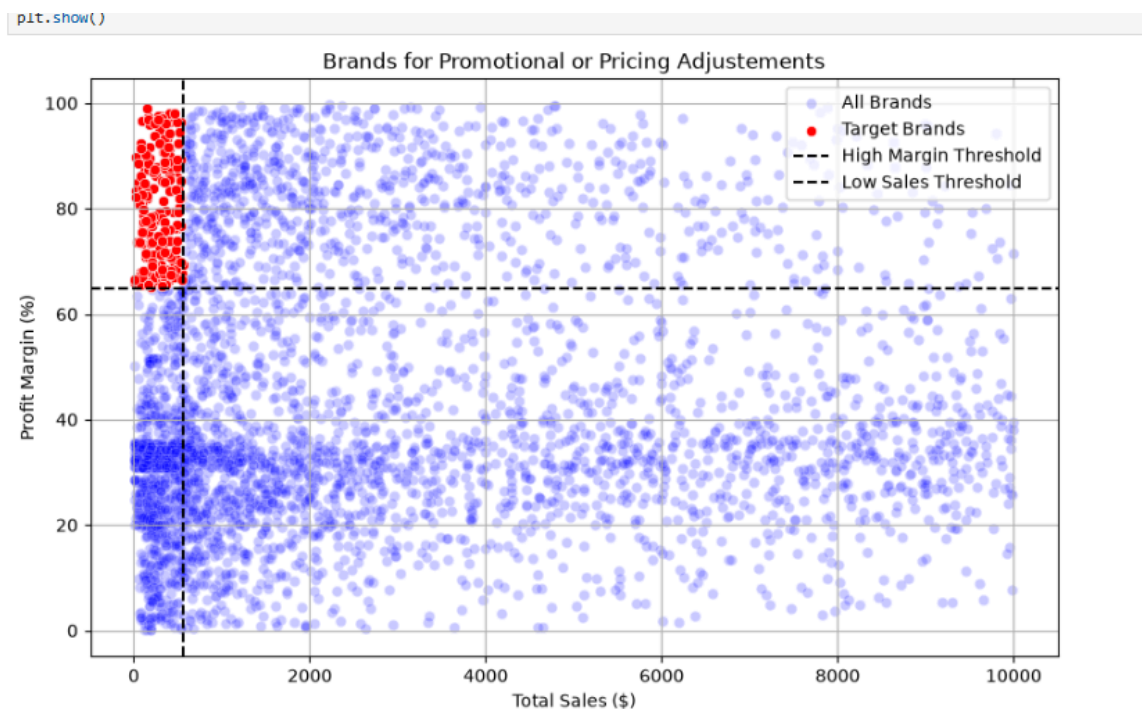


Figure 7.6: Brands Identified for Promotional or Pricing Adjustments

Purpose

- To identify brands that have high profit margins but relatively low sales performance.
- To highlight products that may benefit from promotional campaigns or pricing adjustments.
- To support data-driven marketing and pricing decisions.

Key Observations

- The scatter plot classifies brands based on Total Sales and Profit Margin.
- The red data points represent brands with profit margins above 65% and sales below the defined threshold, making them the target brands for further evaluation.
- Most brands are distributed across different sales and profit margin levels, while only a smaller subset falls within the high-margin, low-sales region.

Key Insights

- These target brands are profitable but are not generating sufficient sales revenue.
- Improving the visibility of these products through promotional campaigns, revised pricing strategies, or targeted marketing could increase sales while maintaining healthy profit margins.
- This analysis helps businesses prioritize marketing efforts toward products with the highest potential return on investment instead of applying promotions across all brands.

Research Question 2

Which vendors and brands demonstrate the highest sales performance?

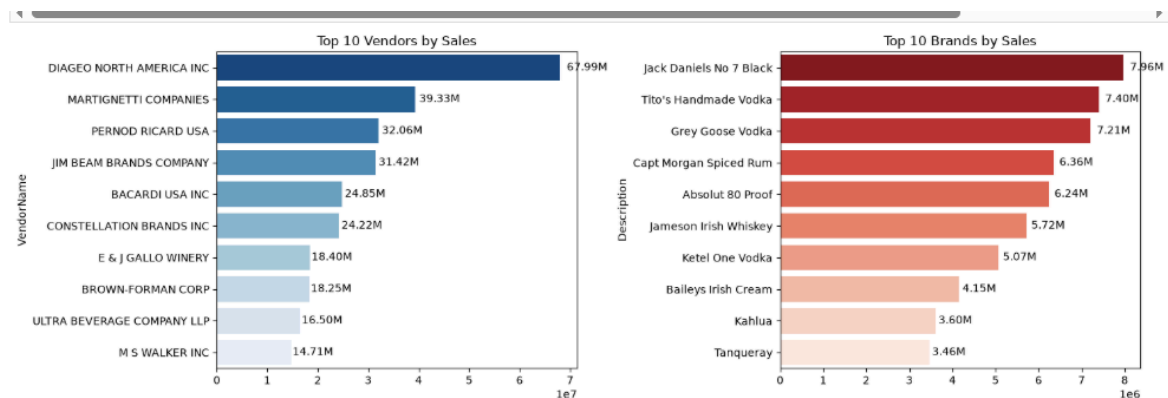


Figure 7.7: Top 10 Vendors and Top 10 Brands by Sales

Purpose

- To identify the highest-performing vendors and brands based on total sales revenue.
- To recognize the key contributors to the organization's overall sales performance.
- To support procurement, inventory planning, and strategic business decisions.

Key Observations

- DIAGEO NORTH AMERICA INC is the highest-performing vendor, generating approximately \$67.99M in total sales, followed by MARTIGNETTI COMPANIES (\$39.33M) and PERNOD RICARD USA (\$32.06M).
- Among the brands, Jack Daniels No. 7 Black records the highest sales at approximately \$7.96M, followed by Tito's Handmade Vodka (\$7.40M) and Grey Goose Vodka (\$7.21M).
- The sales distribution indicates that a limited number of vendors and brands contribute a significant share of the overall revenue.

Key Insights

- The business relies heavily on a few high-performing vendors and brands for revenue generation.
- Maintaining strong relationships with these vendors and ensuring adequate inventory of top-selling brands can help sustain sales performance.

- These top-performing vendors and brands can also serve as benchmarks for evaluating and improving the performance of lower-performing vendors and products.

Research Question 3

Which vendor contribute the most to total purchase dollars?

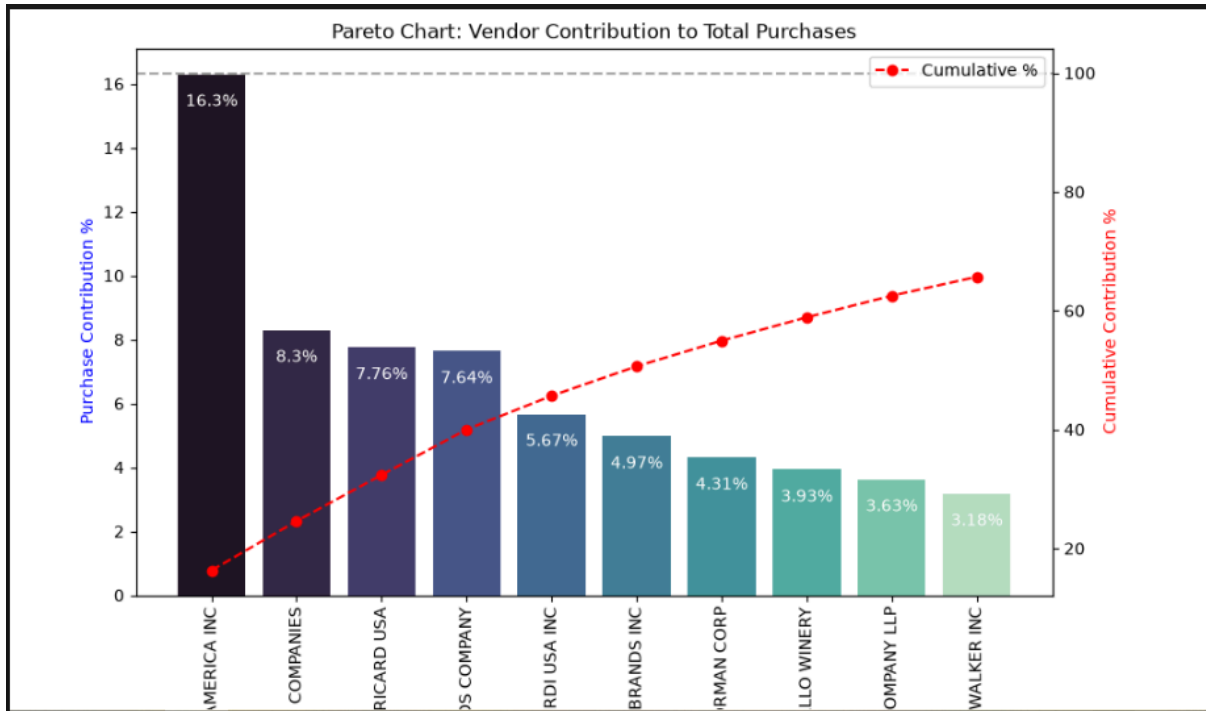


Figure 7.8: Pareto Analysis of Vendor Contribution to Total Purchase Dollars

Purpose

- To identify the vendors contributing the highest share of total purchase dollars.
- To determine whether purchase spending is concentrated among a few key vendors.
- To support vendor prioritization and procurement planning using the Pareto principle.

Key Observations

- DIAGEO NORTH AMERICA INC is the largest contributor, accounting for 16.3% of the total purchase dollars.
- MARTIGNETTI COMPANIES (8.3%), PERNOD RICARD USA (7.76%), and JIM BEAM BRANDS COMPANY (7.64%) are the next major contributors.
- The cumulative contribution line shows that a relatively small group of vendors accounts for a significant portion of the organization's total procurement expenditure.

Key Insights

- Purchase spending is concentrated among a limited number of vendors, indicating a dependency on key suppliers.

- These high-contributing vendors should be prioritized for strategic partnerships, supplier performance monitoring, and contract negotiations.
- Diversifying procurement across additional reliable vendors can help reduce supply chain risks while maintaining operational efficiency.

Interpretation: The Pareto analysis demonstrates that a relatively small number of vendors contribute a large proportion of the total purchase value, supporting the principle that procurement efforts should focus on the most influential suppliers.

Research Question 4

How much total procurement is dependent on the top vendors?

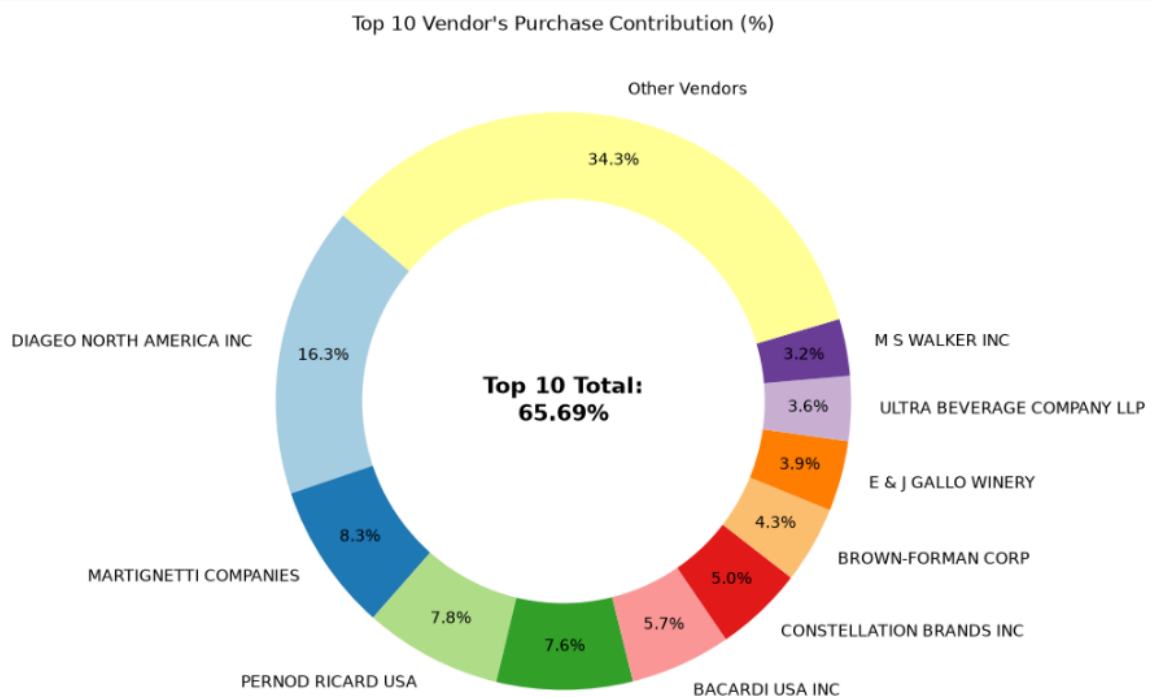


Figure 7.9: Purchase Contribution of the Top 10 Vendors

Purpose

- To determine the overall procurement dependency on the top 10 vendors.
- To measure the percentage contribution of each major vendor toward the total purchase value.
- To evaluate supplier concentration and procurement risk.

Key Observations

- The top 10 vendors collectively contribute 65.69% of the total purchase dollars.
- DIAGEO NORTH AMERICA INC is the largest contributor, accounting for 16.3% of total purchases.
- The remaining vendors outside the top 10 contribute 34.31%, indicating a more distributed contribution among smaller suppliers.

Key Insights

- The organization relies significantly on its top 10 vendors, with nearly two-thirds of procurement concentrated among them.
- Maintaining strong relationships with these key vendors is essential for ensuring a stable and efficient supply chain.
- At the same time, reducing over-dependence on a limited number of suppliers through vendor diversification can improve supply chain resilience and reduce procurement risk.

Research Question 5

Does purchasing in bulk reduce the unit price , and what is the optimal purchase volume for cost savings?

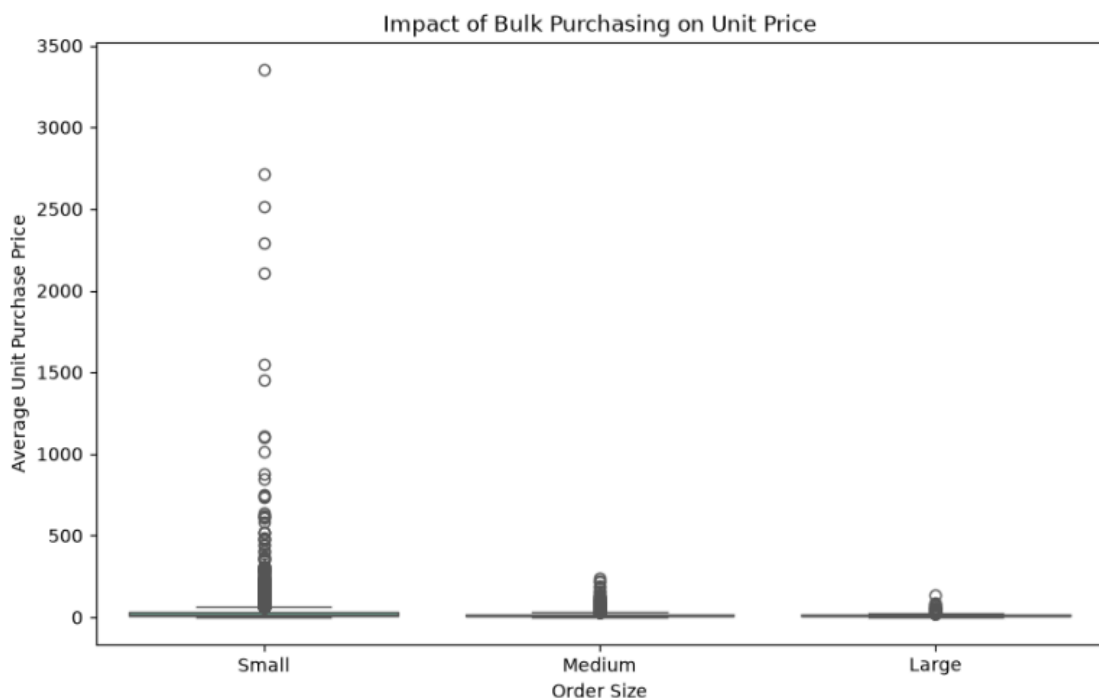


Figure 7.10: Impact of Bulk Purchasing on Unit Purchase Price

Purpose

- To analyze the relationship between order size and average unit purchase price.
- To evaluate whether bulk purchasing leads to lower procurement costs.
- To identify the purchase volume that offers the greatest cost efficiency.

Key Observations

- Small orders exhibit the highest variability in unit purchase prices, with several high-value outliers.
- Medium and large orders show comparatively lower and more consistent unit purchase prices.

- The spread of prices decreases as the order size increases, indicating more stable pricing for bulk purchases.

Key Insights

- The analysis suggests that bulk purchasing is generally associated with lower and more consistent unit purchase prices.
- Medium and large order volumes appear to provide better procurement efficiency compared to small orders.
- Businesses can achieve potential cost savings by consolidating purchases into larger orders, while considering inventory requirements and storage capacity.

Research Question 6

How much capital is locked in unsold inventory per vendor and which vendors contribute the most to it?

Unsold Inventory Value = (Total Purchase Quantity – Total Sales Quantity) × Purchase Price

18]:		
	VendorName	UnsoldInventoryValue
25	DIAGEO NORTH AMERICA INC	722.21K
46	JIM BEAM BRANDS COMPANY	554.67K
68	PERNOD RICARD USA	470.63K
116	WILLIAM GRANT & SONS INC	401.96K
30	E & J GALLO WINERY	228.28K
79	SAZERAC CO INC	198.44K
11	BROWN-FORMAN CORP	177.73K
20	CONSTELLATION BRANDS INC	133.62K
61	MOET HENNESSY USA INC	126.48K
77	REMY COINTREAU USA INC	118.60K

Figure 7.11: Vendor-wise Unsold Inventory Capital

Purpose

- To calculate the amount of capital locked in unsold inventory for each vendor.
- To identify vendors contributing the most to unsold inventory value.
- To support inventory optimization and working capital management.

Key Observations

- The total capital locked in unsold inventory is approximately \$2.71M.
- DIAGEO NORTH AMERICA INC has the highest unsold inventory value (\$722.21K), followed by JIM BEAM BRANDS COMPANY (\$554.67K) and PERNOD RICARD USA (\$470.63K).
- A few vendors account for a significant proportion of the total unsold inventory, indicating higher inventory holding costs.

Key Insights

- High unsold inventory ties up working capital and increases inventory carrying costs.
- Vendors with the highest unsold inventory should be prioritized for inventory optimization through improved demand forecasting, purchasing strategies, or promotional activities.
- Monitoring vendor-wise unsold inventory enables businesses to reduce excess stock, improve inventory turnover, and enhance overall operational efficiency.

Research Question 7

Which vendors have low inventory turnover, indicating excess stock and slow moving products?

] :

VendorName	StockTurnover
ALISA CARR BEVERAGES	0.615385
HIGHLAND WINE MERCHANTS LLC	0.708333
PARK STREET IMPORTS LLC	0.751306
Circa Wines	0.755676
Dunn Wine Brokers	0.766022
CENTEUR IMPORTS LLC	0.773953
SMOKY QUARTZ DISTILLERY LLC	0.783835
TAMWORTH DISTILLING	0.797078
THE IMPORTED GRAPE LLC	0.807569
WALPOLE MTN VIEW WINERY	0.820548

Figure 7.12: Vendors with the Lowest Stock Turnover

Purpose

- To identify vendors with the lowest inventory turnover.
- To detect slow-moving inventory and potential excess stock.

- To support inventory optimization and purchasing decisions.

Key Observations

- ALISA CARR BEVERAGES has the lowest stock turnover (0.615), followed by HIGHLAND WINE MERCHANTS LLC (0.708) and PARK STREET IMPORTS LLC (0.751).
- The listed vendors exhibit relatively low stock turnover values, indicating that their inventory is sold at a slower rate compared to others.
- Lower turnover values suggest that products supplied by these vendors remain in inventory for longer periods.

Key Insights

- Low stock turnover may indicate excess inventory, lower customer demand, or inefficient inventory management.
- Businesses should review purchasing patterns, demand forecasts, and inventory levels for these vendors to minimize inventory holding costs.
- Targeted promotional campaigns or optimized procurement strategies can help improve inventory movement and reduce capital tied up in slow-moving stock.

Interpretation: Vendors with lower stock turnover require closer inventory monitoring, as prolonged stock holding can increase storage costs and reduce overall inventory efficiency

Research Question 8

How do the 95% confidence intervals of profit margins differ between top-performing and low-performing vendors?

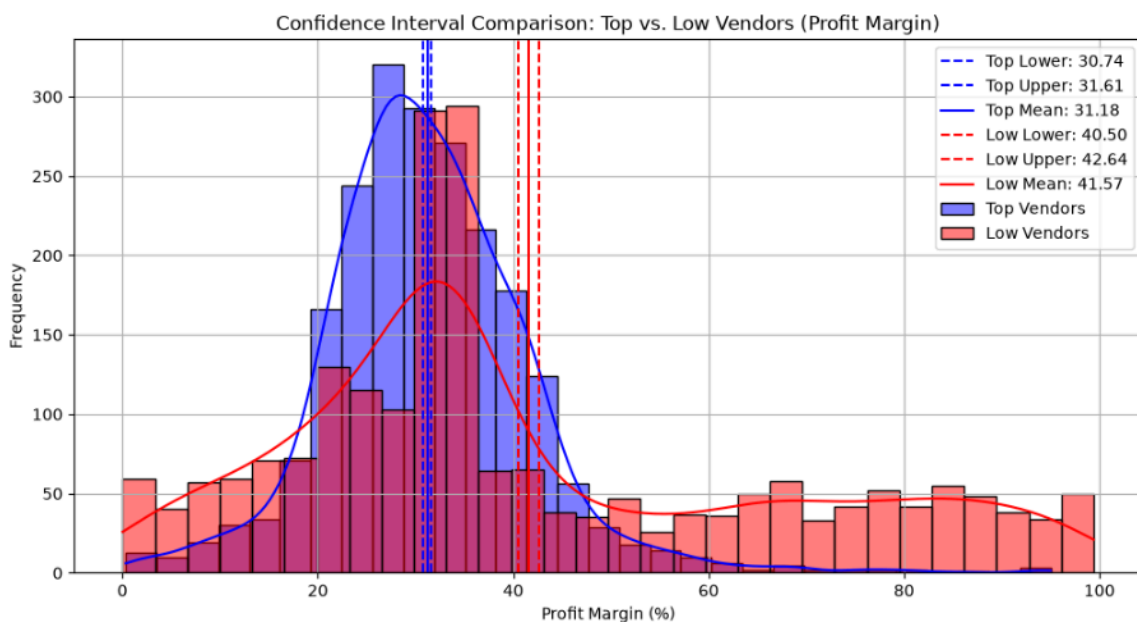


Figure 7.13: 95% Confidence Interval Comparison of Profit Margins

Purpose

- To compare the profit margins of top-performing and low-performing vendors using 95% confidence intervals.
- To estimate the range within which the true average profit margin is likely to lie for each vendor group.
- To assess whether there is a significant difference in profitability between the two groups.

Key Observations

- The top-performing vendors have an estimated mean profit margin of 31.18%, with a 95% confidence interval ranging from 30.74% to 31.61%.
- The low-performing vendors have a higher estimated mean profit margin of 41.57%, with a 95% confidence interval ranging from 40.50% to 42.64%.
- The confidence intervals do not overlap, indicating a clear difference between the average profit margins of the two vendor groups.

Key Insights

- Low-performing vendors achieve higher profit margins despite generating lower sales, suggesting that higher profitability does not necessarily translate into stronger sales performance.
- This finding highlights potential opportunities to improve the sales of high-margin vendors through targeted promotions, pricing strategies, or increased product visibility.
- Confidence interval analysis provides statistical evidence that the observed difference in profit margins is meaningful rather than being due to random variation.

Research Question 9

Is there a significant difference in profit margins between top-performing and low-performing vendors?

Hypothesis:

H0(Null Hypothesis): There is no significant difference in the mean profit margins of top-performing and low-performing vendors.

H1(Alternate Hypothesis): The mean profit margins of top-performing and low-performing vendors are significantly different.

T-Statistics:-17.669523,p-value:0.000000

Reject H0:There is a significant difference in profit margins between top and low performing vendors.

Figure 7.14: Independent Two-Sample *t*-Test for Profit Margin Comparison

Purpose

- To determine whether a statistically significant difference exists between the profit margins of top-performing and low-performing vendors.

- To validate whether the observed difference in profit margins is supported by statistical evidence.

Key Observations

- The analysis compares the profit margins of vendors in the top 25% and bottom 25% based on total sales.
- The independent two-sample *t*-test produced a *t*-statistic of -17.67 with a *p*-value < 0.001.
- Since the *p*-value is significantly lower than the significance level of 0.05, the null hypothesis (H_0) is rejected.

Key Insights

- There is a statistically significant difference in the average profit margins of top-performing and low-performing vendors.
- The variation in profit margins is unlikely to have occurred by random chance, indicating that vendor performance is associated with meaningful differences in profitability.
- These findings support the need for different business strategies for top-performing and low-performing vendors, such as targeted pricing, promotional campaigns, or vendor-specific performance improvement initiatives.

Hypothesis Tested

- **H_0 (Null Hypothesis):** There is no significant difference in the mean profit margins of top-performing and low-performing vendors.
- **H_1 (Alternative Hypothesis):** There is a significant difference in the mean profit margins of top-performing and low-performing vendors.

Decision: Since *p*-value < 0.05, H_0 is rejected and H_1 is accepted.

Research Question Summary

The research questions addressed in this chapter provided valuable insights into vendor performance, purchasing behavior, pricing strategies, inventory management, and profitability. Through statistical analysis and data visualization, several important business findings were identified:

- Brands with high profit margins but lower sales were recognized as potential candidates for promotional and pricing strategies.
- A small number of vendors and brands were found to contribute a significant share of the organization's total sales and purchases.
- Procurement analysis revealed that nearly two-thirds of the total purchase value is concentrated among the top vendors, highlighting supplier dependency.
- Inventory analysis identified vendors with low stock turnover and high unsold inventory value, indicating opportunities to optimize inventory management.

- Statistical hypothesis testing confirmed a significant difference in profit margins between top-performing and low-performing vendors, providing evidence for differentiated business strategies.

Overall, these findings demonstrate how data-driven analysis can support better decision-making in vendor management, procurement planning, pricing optimization, and inventory control. The insights obtained from this analysis also served as the foundation for designing the interactive Power BI dashboard presented in the next chapter.

CHAPTER 8

Interactive Power BI Dashboard

8.1 Introduction

After completing the ETL pipeline, feature engineering, exploratory data analysis, and statistical analysis, an interactive dashboard was developed using Microsoft Power BI. The dashboard integrates key performance indicators (KPIs), dynamic filters, and business visualizations into a single interface, enabling users to monitor vendor performance, procurement trends, sales performance, profitability, and inventory metrics.

The dashboard is designed to support data-driven decision-making by allowing users to interactively explore vendor and product performance through slicers, KPI cards, and multiple analytical visualizations.

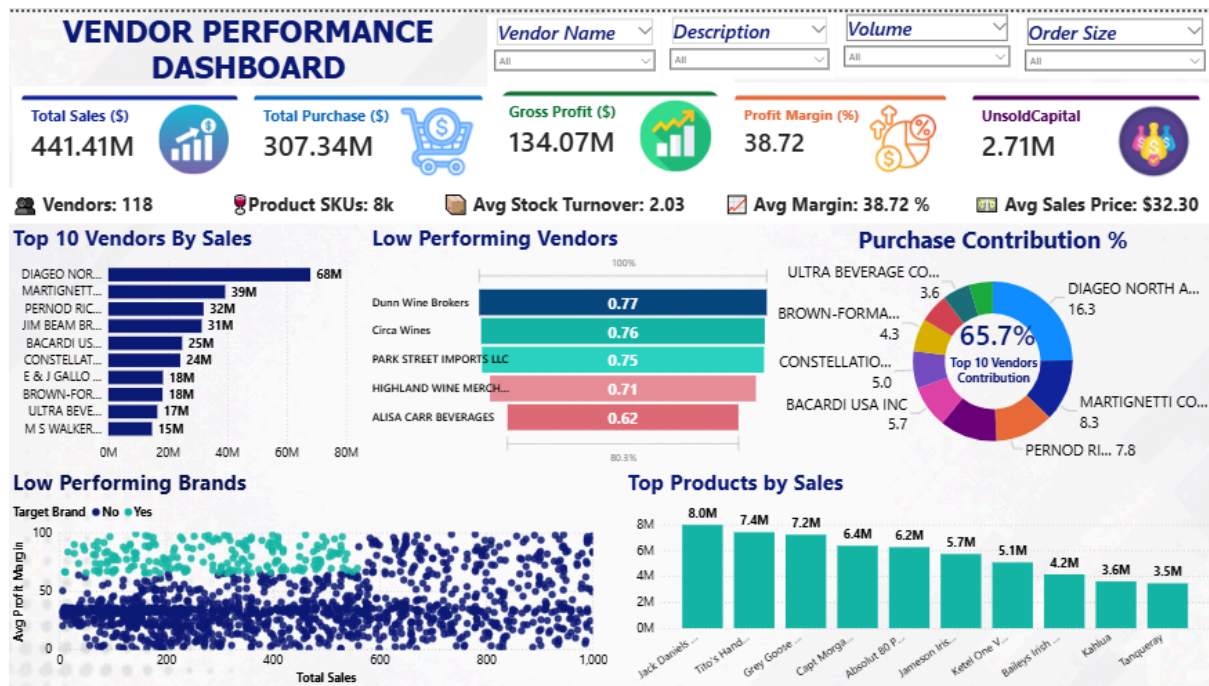
8.2 Dashboard Overview

The Vendor Performance Dashboard provides a comprehensive view of procurement, sales, profitability, and inventory performance. It combines summary KPIs with interactive visualizations, enabling users to analyze vendor performance from multiple business perspectives.

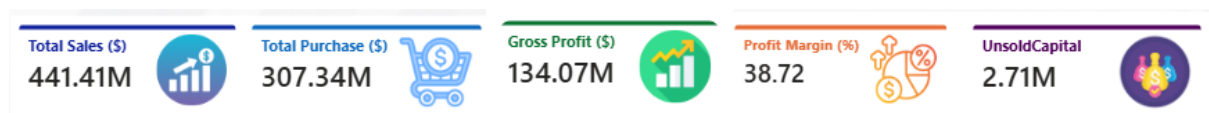
The dashboard consists of:

- KPI Cards for key business metrics.
- Interactive slicers for dynamic filtering.
- Vendor and brand performance visualizations.
- Purchase contribution analysis.
- Inventory and stock turnover analysis.
- Product sales analysis.

The dashboard updates dynamically based on user-selected filters, providing real-time insights into vendor and product performance.



8.2.1 KPI Cards



The KPI (Key Performance Indicator) cards provide a quick summary of the organization's overall business performance.

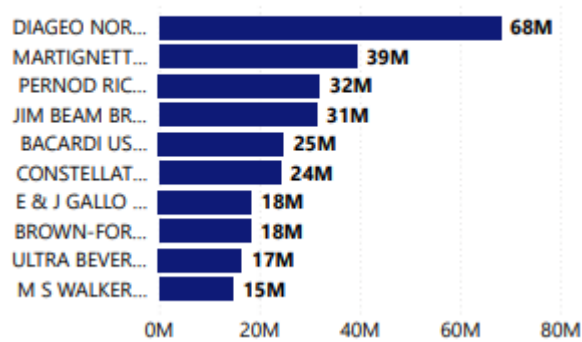
The dashboard includes five KPI cards:

- Total Sales (\$): Displays the total revenue generated from product sales.
- Total Purchase (\$): Represents the total procurement cost incurred for purchasing inventory.
- Gross Profit (\$): Shows the overall profit earned after deducting purchase costs from sales revenue.
- Profit Margin (%): Indicates the percentage of profit earned relative to total sales, helping evaluate overall profitability.
- Unsold Capital: Represents the monetary value of inventory that remains unsold, highlighting the amount of working capital tied up in stock.

These KPI cards are calculated using DAX measures and automatically update based on the selected slicers, allowing users to monitor key business metrics dynamically across different vendors, products, volumes, and order sizes.

8.2.3 Top 10 Vendors by Sales

Top 10 Vendors By Sales

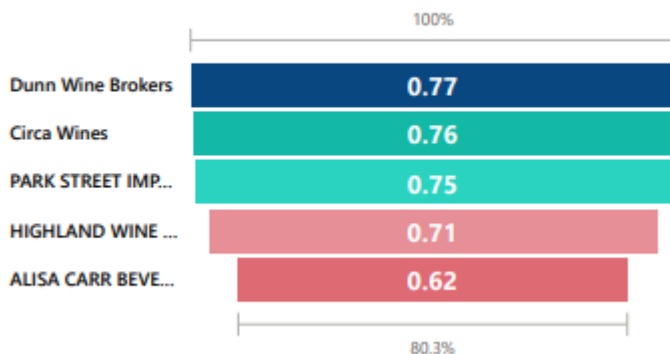


The Top 10 Vendors by Sales visualization is displayed using a horizontal bar chart to compare the sales performance of the highest-performing vendors. This chart provides a clear ranking of vendors based on total sales, making it easy to identify the organization's major revenue contributors.

A horizontal bar chart was selected because it offers better readability for vendor names and enables quick comparison of sales values across multiple vendors. The visualization responds dynamically to the dashboard slicers, allowing users to analyze vendor performance for specific products, volumes, or order sizes. This helps procurement managers and business analysts monitor vendor performance and identify key suppliers efficiently.

8.2.4 Low Performing Vendors

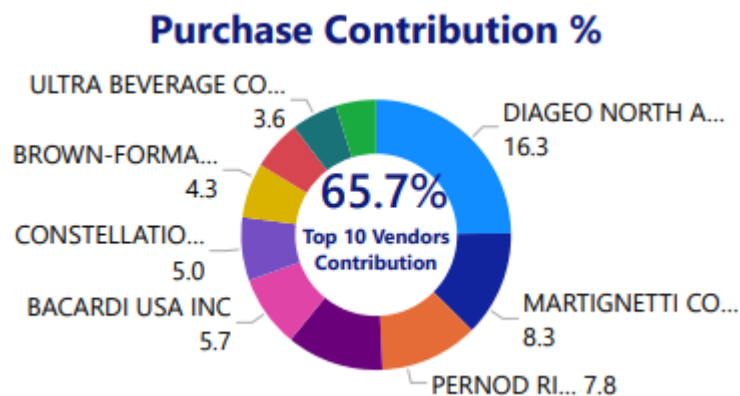
Low Performing Vendors



The Low Performing Vendors visualization highlights vendors with the lowest stock turnover values, helping users identify suppliers associated with slow-moving inventory. Displaying this information in a horizontal bar chart allows users to quickly compare vendor performance and recognize vendors requiring further investigation.

The visualization is fully interactive and updates automatically based on slicer selections. It supports inventory management by enabling decision-makers to identify vendors with lower inventory efficiency and prioritize actions such as inventory optimization, purchasing adjustments, or demand planning.

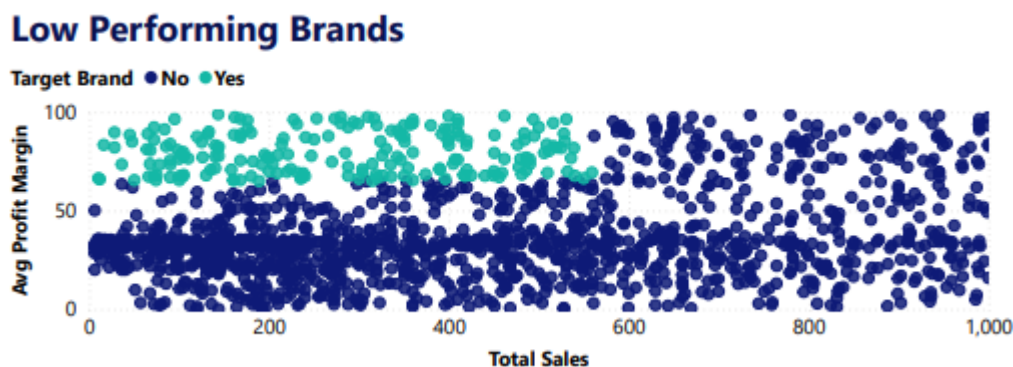
8.2.5 Purchase Contribution



The Purchase Contribution visualization is represented using a donut chart to illustrate the percentage contribution of the top vendors toward the organization's total purchase value. The chart provides an intuitive view of procurement concentration while emphasizing the cumulative contribution of the leading vendors.

The donut chart enables users to understand supplier dependency at a glance and supports procurement planning by identifying vendors that account for a significant share of purchase expenditure. The visualization updates dynamically with the applied dashboard filters, allowing users to analyze purchase contributions across different business scenarios.

8.2.6 Low Performing Brands

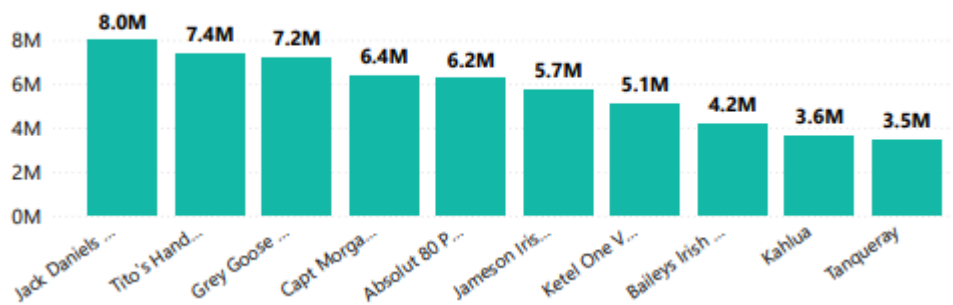


The Low Performing Brands visualization is presented using a scatter plot to analyze the relationship between Total Sales and Average Profit Margin. This chart helps identify brands with lower sales performance while distinguishing those that maintain relatively high profit margins.

A scatter plot was chosen because it effectively displays the distribution of brands across two business metrics simultaneously, making it easier to identify target brands for promotional campaigns or pricing adjustments. The visualization updates dynamically based on the selected slicers, enabling users to analyze brand performance under different business conditions.

8.2.7 Top Products by Sales

Top Products by Sales



The Top Products by Sales visualization is displayed using a column chart to compare the sales performance of the highest-selling products. The chart provides a clear comparison of product-wise sales, allowing users to identify products that contribute the most to overall revenue.

A column chart was selected because it enables straightforward comparison of sales values across different products while maintaining readability. The visualization responds to all dashboard filters, allowing users to explore product performance for selected vendors, volumes, and order sizes. This supports inventory planning, product performance evaluation, and demand analysis.

8.3 Dashboard Features

The Vendor Performance Dashboard incorporates several interactive features to enhance usability and support business analysis. Key features include:

- Dynamic KPI cards that provide an instant overview of key business metrics.
- Interactive slicers for filtering data by vendor, product description, volume, and order size.
- Automatic cross-filtering, where all visualizations update based on user selections.
- Multiple visualization types, including bar charts, column charts, donut charts, and scatter plots, to represent different aspects of vendor performance.
- A single-page dashboard layout that enables users to monitor sales, procurement, profitability, inventory, and product performance from one interface.
- DAX-based calculations that ensure business metrics are updated dynamically according to applied filters.

CHAPTER 9

Results and Business Recommendations

9.1 Key Findings

The project successfully analyzed vendor performance by combining ETL processing, feature engineering, statistical analysis, and interactive dashboarding. The major findings are summarized below:

- The top 10 vendors contribute 65.69% of the total purchase value, indicating a high dependency on key suppliers.
- Several brands were identified with high profit margins but relatively low sales, making them suitable candidates for promotional strategies.
- Vendors with low stock turnover and high unsold inventory were identified, highlighting opportunities for inventory optimization.
- Statistical analysis confirmed a significant difference in profit margins between top-performing and low-performing vendors.
- The interactive Power BI dashboard enables dynamic monitoring of sales, procurement, profitability, and inventory performance.

9.2 Business Recommendations

Based on the analysis, the following recommendations are proposed:

- Strengthen relationships with high-performing vendors.
- Promote high-margin brands with lower sales.
- Optimize procurement for vendors with low inventory turnover.
- Monitor unsold inventory to reduce working capital blockage.
- Use the Power BI dashboard for continuous performance monitoring and data-driven decision-making.

9.3 Chapter Summary

This chapter summarized the key findings obtained from the vendor performance analysis and presented practical business recommendations. The insights generated through this project can support procurement planning, inventory optimization, and strategic vendor management while enabling informed business decisions through the interactive dashboard.